# Photovoltaic Connector Cross-Mating and Compatibility - Technical Warning

## 1. What cross-mating is

Many connectors are sold as 'MC4-compatible' or 'MC4-type'. These are not the genuine part and are not certified to mate with it or with each other. Forcing two different makes together is 'cross-mating' - a hidden and widespread field defect that passes day-one inspection and fails later.

## 2. Why it is hidden [p2]

Cross-mated halves latch, pass a day-one visual and even an initial flash / Isc test, but they do not achieve the designed contact normal-force and geometry. The result is a high-resistance, unstable contact that heats under load and melts weeks to months later. Failures cluster in the first hot, high-irradiance season after commissioning, which is why they look like 'random' connector burns.

## 3. Mechanism

- mismatched contact-spring geometry -> lower or uneven contact force
- -> higher, unstable contact resistance
- -> localised I2R heating and oxidation
- -> thermal runaway -> melt and DC series arc

## 4. Standards

- IEC 62852 - connectors shall be mated only with the same type from the same manufacturer, unless a specific pairing is certified as intermateable.
- UL 6703 - covers the intermateability evaluation.
- Mixing uncertified brands voids the connector rating and frequently the module / inverter warranty.

## 5. How to identify in the field

**Step What to check**

- Markings | brand + part number moulded on BOTH halves
- Match | are both halves the same make and series?
- Thermal | IR-scan under load for connectors hotter than neighbours
- Resistance | micro-ohm / contact-resistance check if a meter is available

## 6. Remediation

1. If the halves are mismatched, cut out and replace BOTH with a matched genuine pair.
2. Do not 'make do' by re-latching the existing mismatched pair.
3. Document the make and part number used for the repair.

## 7. Why this is a FLEET issue, not a one-off

Cross-mating is usually an install-crew habit or a procurement substitution, so the SAME defect is repeated across many strings and arrays built by the same crew or from the same lot. A single found cross-mated connector should trigger an array-wide audit and a same-installer fleet-wide review. This is exactly where a blast-radius / Fleet agent escalates one field finding into a systemic, money-saving review.

## 8. Procurement and QA guidance

- Specify a single approved connector make/type per project; control it on the BOM.
- Reject field substitutions; verify make at goods-in.
- Check connector make and IR-signature at commissioning (IEC 62446).